# BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2024

# BIOLOGY

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

**Maximum Marks:** 12

Note: Note: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Mark.

Q1. 1. Choose the correct answer from the given options:

Q1. Scientific name of frog is:

(A) Palaeon
(B) Rana tigrina
(C) Periplaneta
(D) Pheretima

Q2. The cell organelle that is found in animal's cells :and helps in Intracellular Intracellular digestion is called:

(A) Ribosome
(B) lysosome
(C) Mitochandria
(D) Golgi bodies

Q3. Stage of mitosis in which chromatids reach the poles and their movement ceases, is called:

(A) Prophase
(B) Metaphase
(C) Anaphase
(D) Telophase

Q4. Biological method involves all of the following except:

(A) Data collection
(B) Observation
(C) Experiment
(D) Proportion

Q5. Fundamental molecule produced during photosynthesis is. :

(A) Glucose
(B) Fatty acid
(C) Amino acid
(D) Nucleotide

Q6. In human digestive system, trachea and esophagus both are : connected to:

(A) Stomach
(B) Large Intestine
(C) Pharynx
(D) Larynx

Q7. Splitting of water presence of sun light, is called:

(A) Hydrolysis
(B) Glycolysis
(C) Photolysis
(D) None of these

Q8. It is involved in the clotting of blood:

(A) Red Blood Cells
(B) White Blood Cells
(C) Platelets
(D) Haemoglobin

Q9. Part of a root between epidermis and endodermis is called:

(A) Xylem
(B) Phloem
(C) Comblum
(D) Cortex

Q10. Volvox is a poly phyletic genus of:

(A) Green Algae
(B) Red Algae
(C) Brown Algae
(D) None of these

Q11. Chemical reaction to carry down at proper rate, conditions are : required:

(A) Temperature and nature
(B) Nature and pressure
(C) Nature and stracture
(D) Temperature and pressure

Q12. Stage of aerobic respiration takes place at the cristae of mitochondria is called

(A) Electron transport chain
(B) Glycolysis
(C) Krebs's cycle
(D) C3 cycle C3

## SECTION “B” : (Short Answer Question) Note: Attempt any EIGHT questions from this section. Each question carries Three

**Maximum Marks:** 24

Q2. Define the following: (iii) (ii) (i) :

(ii) Environmental

(i) Parasitology

(iii) Biotechnology Biology

Q3. Explain the biological method with the help of a flowchart.

Q4. Describe any three principles of classification.

Q5. Write any three differences between light microscope and electron microscope.

Q6. Write different types of chromosomes with their diagrams.

Q7. Define mitosis

Q8. Write any three effects of malnutrition on human.

Q9. Draw a neat and labelled diagram of “Amoeba”.

Q10. Describe any three causes of deforestation.

Q11. Write down any one function each of the following: (iii) (ii) (i) :

(i) Mitochondria

(ii) Golgi Bodies

(iii) Plastids

Q12. Write down any three characteristics of kingdom Fungi.

Q13. “Careers in Biology”. Describe any three fields of it. " "

## SECTION “C” : (Detailed –Answer Question)

**Maximum Marks:** 24

> Note: Note: Attempt any Four questions from this Section. Each question carries SIX (6) marks.

Q14. Draw a neat and labelled diagram of Human Alimentary canal OR internal structure of Human Heart.

Q15. Write the name of two blood diseases. Describe their symptoms and causes.

Q16. Define the following terms: (vi) (v) (iv) (iii) ( ) (ii) (i) :

(i) Vitamins

(ii) Goiter

(iii) Stomata

(iv) Cisternae

(iv) Nucleoplasm

(vi) Activator

Q17. What are the major components of human food? Describe any two components.

Q18. Define enzyme. Write any four characteristics of it.

Q19. Write any six differences between respiration and photosynthesis in plants.